

SQL Review

Quiz Format

Closed book. 90 minutes, 100 points.

- **Part 1:** Multiple choice & short answer
 - 30 minutes, 20 questions × 2 points = 40 points
- **Part 2:** Write `SELECT` queries
 - 60 minutes, 12 questions × 5 points = 60 points
 - Database schema and sample data provided

Quiz Prep Tips

1. **Scope:** slides and in-class code examples only. DataCamp content is not tested unless covered in class.
2. **Practice writing queries from scratch**, closed book. Reading a query and writing one are different skills.
3. **Take the mock exam on Ed** under quiz conditions: timed, no notes.
4. **Post questions on Ed**, so everyone sees the answer.

Basic SELECT Syntax

SELECT all columns and rows from person table

```
SELECT * FROM person;
```

Limit number of rows returned

```
SELECT * FROM person LIMIT 5;
```

Order rows by a specific column

```
SELECT * FROM person ORDER BY name;  
SELECT * FROM person ORDER BY id DESC;
```

Select specific columns from person table

```
SELECT id, name FROM person;
```

Basic SELECT Syntax

Select specific rows from person table

```
SELECT * FROM person WHERE id = 1;  
SELECT * FROM person WHERE age > 30;  
SELECT * FROM person WHERE age BETWEEN 20 AND 30;
```

Select rows with pattern matching

```
SELECT * FROM person WHERE name LIKE 'Alice%';  
SELECT * FROM person WHERE id LIKE '_H42W%';
```

Advanced SELECT Syntax

Aggregate functions

```
SELECT COUNT(*) FROM person;  
SELECT MIN(age) FROM person;  
SELECT MAX(age) FROM person;
```

Aggregate functions and GROUP BY

```
SELECT person_id, COUNT(*) AS checkin_count  
FROM facebook_event_checkin GROUP BY person_id;
```

Having clause

```
SELECT person_id  
FROM facebook_event_checkin GROUP BY person_id  
HAVING COUNT(*) > 1;
```

Advanced SELECT Syntax

Subquery to reuse query results

```
SELECT id, name, ssn
FROM person
WHERE license_id IN (
    SELECT id FROM driver_license WHERE hair_color = 'red');
```

JOINS to combine tables

```
-- without table aliases
SELECT person.id, person.name, drivers_license.hair_color
FROM person JOIN drivers_license ON person.license_id = drivers_license.id
WHERE drivers_license.hair_color = 'red';
-- with table aliases
SELECT p.id, p.name, dl.hair_color
FROM person AS p JOIN drivers_license AS dl ON p.license_id = dl.id
WHERE dl.hair_color = 'red';
```

DDL & DML to Create and Modify Tables

```
DROP TABLE IF EXISTS movies;
CREATE TABLE movies (
    id INT,
    main_char_id INT,
    title text
);

ALTER TABLE movies ADD COLUMN year INT;
ALTER TABLE movies DROP COLUMN year;

ALTER TABLE movies ADD CONSTRAINT pk_movies PRIMARY KEY (id);

ALTER TABLE movies
ADD CONSTRAINT fk_main_char FOREIGN KEY (main_char_id) REFERENCES students(id);
```

DDL & DML to Create and Modify Tables

```
-- continued from previous slide
INSERT INTO movies (id, main_char_id, title)
VALUES (1, 101, 'The Great Adventure'),
       (2, 102, 'Another Story');

UPDATE movies SET title = 'The Greatest Adventure' WHERE id = 1;

DELETE FROM movies WHERE id = 1;

-- FK constraint violation if 103 not in students.id
INSERT INTO movies VALUES (3, 103, 'New Movie');

DROP TABLE movies;
```